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SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Duration: 60 Minutes

Total Marks: 25

Code of Conduct (192525122)

I K. Mahesh Babu (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K. Mahesh Babu

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty

Total Marks Awarded:

Describe the Evolution from standalone computing to cloud based Systems. Summarize key technologies transitions that enabled cloud computing.

cloud Computing evolved through different Stages:

1. Standalone Computing.

- A computer works independently.
- Data and Software are stored on the same computer.
- No internet connection is required.

2. Client-server Computing:

- Multiple server computers work together to complete task.
- Reserver stores data and provides services.

3. Distributed Computing:

- Multiple computers work together to complete task.
- Resources are shared through a network.

4. cloud Computing:

- Computing resources such as servers, storage and software are provided over the internet.
- users can access services anytime and from anywhere.

Key Technologies:-

- High-speed Internet.
- Virtualization.
- Data Centers.
- Web Services and APIs
- Distributed Computing.

Q. Interpret the working of hypervisors in Virtualization.
Differentiate between Type1 and Type2 hypervisors with Examples.

A. A Hypervisor is Software that creates and manages virtual machine (VMs). It allows multiple operating systems to run on one physical computer.

Working:-

- Divides CPU, RAM, storage and network resources.
- Creates separate virtual machines.
- Each VM runs independently.

Difference b/w Type1 and Type2:-

Type1 Hypervisor:-

- Runs directly on hardware.
- faster Performance.
- More secure.
- used in cloud data centers.

Ex: VMware ESXi, Microsoft Hyper-V etc.

Type 2:

- Runs on host operating system.
- Slightly slower.
- less secure.
- used for Personal Computers.

• Examples:

Oracle Virtual Box, VMware Workstation.

Example:

VMware ESXi is used by cloud providers to run many virtual machines on one server.

3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between the services.

A An API (Application programming interface) is set of rules that allows different applications to communicate and exchange data with each other. In cloud computing APIs are essential because they enable different cloud services, applications, and platforms to work together efficiently.

Role of APIs & Cloud Service:

- APIs connect different cloud services and applications.
- They allow applications to exchange data securely over the internet.
- APIs help automate cloud operations such as creating virtual machines, managing stores and monitoring resources.
- They allow developers to access cloud services without knowing the internal implementation.
- APIs support integration between cloud providers and third-party applications.

Benefits of APIs:

- faster application development
- Easy integration of multiple cloud services.
- Improved automation and efficiency.
- Better scalability and flexibility.
- Secure data exchange using authentication methods.

How APIs Support Interoperability:

- Providing a standard communication interface.
- Enabling data sharing between different cloud services.

- Allowing applications built with different of programming languages to interact.
- Supporting integration between public, private, and hybrid cloud environments.

Examples:

- Google Maps API is used in food delivery apps to show customer locations.
- Payment APIs: (Such as Razorpay or Paypal) allow websites to process online payments.
- Google Drive API: allows applications to upload, download, and manages files stored in the cloud.

Conclusion:

APIs are the backbone of cloud service integration. They enable secure communication, seamless data sharing, and interoperability b/w different cloud services, making cloud applications more flexible, scalable and efficient.

4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.

Resource pooling is one of the essential characteristics of cloud computing. It means that a cloud services provider combines physical storage such as servers, storage, memory, processing power, and networking into a shared pool. These resources are dynamically allocated and relocated to multiple users based on their requirements.

How resources pooling works:

- Multiple users share the same cloud infrastructure.
- Resources are assigned according to user demand.
- When demand decreases, the resources are returned to the shared pool.
- The cloud provider manages resource allocation automatically.

Benefits of Resource Pooling:

- Efficient utilization of computing resources.
- Reduces infrastructure and maintenance costs.
- Supports multiple users simultaneously (multi-tenancy)
- Provides high scalability as user demand changes.
- Improves overall system performance.

Example:

A cloud storage devices like Google drive (G) Microsoft one drive serves millions of users using the same data center. Each users get their own secure storage space while sharing the same underlying cloud infrastructure.

Conclusion:

Resource Pooling helps cloud providers efficiently share computing resources among the multiple users. It reduces costs, improve performance ensure better resource utilization, and enables cloud services to serve many users simultaneously without affecting security (or) reliability.

5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Rapid elasticity is one of the key characteristics of cloud computing. It refers to the ability of the cloud to quickly increase (or) decrease computing resources (such as CPU, memory, storage, and network bandwidth) based on user demand. Resources can be added (or) removed automatically without interrupting the service.

How Rapid Elasticity supports Dynamic workloads:

- Automatically increases resources during peak demand.
- Reduces resources when demand is low.
- Ensures continuous and smooth application performance.
- Prevents system overload and downtime.
- Supports sudden increases (or) in users or traffic.

Benefits of Rapid Elasticity:

- Cost-effective because users pay only for the resources they use.
- Improves application Performance.
- Provides high scalability and flexibility.
- Ensures better user Experience.
- Reduces manual effort in resource of management.

Example:

An e-commerce website like Amazon receives a huge number of visitors during festive sales (Such as Diwali or Black Friday). The cloud automatically adds more servers to handle the increased traffic. After the same sale ends, the extra servers are removed, reducing costs, while maintaining efficient performance.

Conclusion:

Rapid Elasticity enables cloud system to respond quickly to changing workloads by automatically scaling resources up (&) down. It improves the performance, reduce costs, ensures high availability.

and makes the cloud computing suitable for the applications with varying user demand.

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow